FINITE SECTION LIFTING AND ZARISKI'S MAIN THEOREM: A COMPARISON WITH EGA, HORROCKS, AND THE PESKINE ROUTE

Abstract. The argument in `zmt-proof.tex` gives a constructive proof of the finite-fibre algebraic form of Zariski's Main Theorem. Its conclusion was already proved constructively by Alonso, Coquand and Lombardi. Its cohomological organisation belongs to the formal-functions tradition, but its finite section correction and extraction differ from the Stein-factorisation proof in EGA and from the conductor argument used in Stacks. We identify the precise classical antecedents and the adaptations made for the checked proof. This comparison does not establish historical priority for the particular combination.

1. The statement and the two proof routes

Let R be a commutative ring, I an ideal, and $E = R[x_1, \dots, x_n]$ an algebra. The statement under discussion is

$$E/IE \text{ finite over } R/I \quad \Longrightarrow \quad \exists s \in 1 + IE : \quad B = R[s, sx_1, \dots, sx_n] \text{ finite over } R, \quad B_s \simeq E_s.$$

Here "finite" means finite as a module. There is no noetherian or finite-presentation assumption. This is the conclusion of Alonso–Coquand–Lombardi (ACL), Theorem 1.3 and Corollary 1.4 [1]; their remark also removes injectivity of $R \to E$. Thus neither this statement nor its constructive validity is new.

Peskine, ACL, and Stacks. ACL follows Peskine, using a conductor, strong transcendence and subresultants to replace the minimal-prime argument, then induction on algebra generators ([1], §§2–3, especially Propositions 2.16–2.18 and Theorem 3.1). Stacks §10.123 likewise develops the conductor and strong-transcendence route; its introduction explicitly cites Peskine, Raynaud and notes by Coquand [4]. Our proof does not follow that chain of lemmas. Peskine's original 1966 article was not inspected for this comparison: the detailed comparison is with ACL and the actual Stacks proof.

EGA and the projective route. EGA III_1, §4.3, obtains Stein factorisation from formal functions and coherence of proper direct images. Proposition 4.4.1 identifies the isolated-fibre locus in the proper case; Theorem 4.4.3 applies this after a projective compactification [2]. This is a different route from the one used in Stacks §10.123. The relevant chain is

$$\text{formal functions} \longrightarrow \text{Stein factorisation and connected fibres} \longrightarrow \text{ZMT}.$$

Our argument retains projective charts and section lifting, and replaces this ending by explicit integrality certificates.

2. The exact classical approximation principle

Let W be proper over a noetherian ring A, and $\mathcal{F}$ coherent. For an ideal I and a fixed p, Stacks Lemma 30.20.4(3) gives c such that

$$\begin{aligned} &\operatorname{im}\big(H^p(W, \mathcal{F}/I^k\mathcal{F}) \longrightarrow H^p(W, \mathcal{F}/I^r\mathcal{F})\big) \\ &\qquad = \operatorname{im}\big(H^p(W, \mathcal{F}) \longrightarrow H^p(W, \mathcal{F}/I^r\mathcal{F})\big) \qquad (k \geq r + c). \end{aligned}$$

This statement precedes the completion theorem [6]. Its proof uses finiteness over the Rees algebra of graded cohomology and of its obstruction submodule [5].

This finite control is already explicit in EGA III_1: (4.1.7.12) makes the graded obstruction module finite, (4.1.7.20) kills its sufficiently distant transition maps, and (4.1.7.21) identifies the stable image

Date: 1 November 2026.

with the image of genuine cohomology. The comparison proof is credited there to Serre (footnote on p. 126) [2]. Consequently, replacing completion by a finite approximation step is not itself a new idea.

For the application below, only $p = 0$ and $r = 1$ are needed. One needs an actual section with the prescribed reduction modulo I; one does not need an actual section inducing a prescribed entire compatible formal section. Formal functions identifies a completion, and does not assert that every element of that completion comes from the uncompleted module.

The precise connection with Horrocks. In Theorem 1, pp. 714–715, Horrocks extends a bundle to $\mathbf{P}^1_A$ for a noetherian local ring A. After twisting, the special-fibre splitting has nonnegative degrees and a trivial summand. Vanishing of H^1 gives surjectivity through the infinitesimal thickenings; EGA's formal-functions theorem then lifts a chosen special-fibre section to a genuine section. That section splits off a summand and enables induction on rank [3]. The common mechanism is lifting one useful residual section. Horrocks does not give a ZMT proof here, and our ZMT proof does not assume the analogous H^1 vanishing.

3. A direct mathematical bridge to the certificate

The following explains the comparison mathematically. It is our reconstruction from the approximation principle, rather than an attribution of this exact proof to EGA or Horrocks.

First suppose $R = A$ is noetherian. Put $X = \operatorname{Spec} E$ and choose a projective compactification $X \subset W \subset \mathbf{P}^n_A$, with $X = D_+(Z)$. Write $W_k = W \times_A A/I^k$ and $X_k = X \times_A A/I^k$.

The algebra E/I^kE is finite over A/I^k: choose a monic polynomial P_i for each x_i modulo IE; then $P_i(x_i)^k = 0$ modulo I^kE. The open immersion $X_k \to W_k$ is also proper, because X_k is finite over A/I^k and W_k is separated. Thus X_k is open and closed in W_k. Extend the section Z^{-1} of $\mathcal{O}(-1)$ on X_k by zero to its complement. These sections have the same reduction on W_1.

Apply the approximation principle to $\mathcal{F} = \mathcal{O}_W(-1)$. It gives $\zeta \in H^0(W, \mathcal{O}_W(-1))$ whose restriction to X_1 is Z^{-1}. Therefore

$$s = (Z\zeta)|_X \in E, \qquad s \equiv 1 \pmod{IE}.$$

On each chart $X_i \neq 0$, represent ζ by a homogeneous fraction of degree -1. Comparing it with the fraction on the chart Z and clearing denominators gives

$$sx_i^{a_i+1} = q_i(x), \qquad \deg q_i \leq a_i. \tag{1}$$

The exponents may be increased to clear localisation equalities as well as the displayed denominators. No injection of a chart ring into a further localisation is required. Choose also a polynomial expression $s = \sigma(x)$ with $\deg \sigma \leq D$.

Proposition 3.1. *The data* (1) *imply that $s, sx_1, \ldots, sx_n$ are integral over A.*

Proof. Let F_d be the span of monomials of total degree at most d, and put $d = D + \sum_i a_i$. For a monomial x^β of degree at most $d+1$, if $\beta_i > a_i$ for some i, use (1) to express sx^β with degree at most $|\beta| - 1 \leq d$. Otherwise use $s = \sigma(x)$, obtaining degree at most $D + \sum_i a_i = d$. Hence $sF_{d+1} \subseteq F_d$. The finite module $V = F_d$ contains 1 and is stable under both s and every sx_i. Apply the determinant trick separately to these endomorphisms of the same module V. □

Monomial reduction in the resulting monic equations makes $B = A[s, sx_1, \ldots, sx_n]$ finite. Every x_i is $(sx_i)/s$ in E_s, and localisation preserves the injection $B \subseteq E$; hence $B_s \simeq E_s$.

This ending explains the purpose of the negative twist: a denominator of exponent $a_i + 1$ has a numerator of degree a_i, giving exactly the one-degree drop in (1). It needs one finite stable module, without constructing the finite algebra $H^0(W, \mathcal{O}_W)$ or proving connectedness of fibres.

4. What the constructive proof changes

The checked proof develops the preceding mechanism using finite polynomial and Laurent data. It first extracts finitely many residual monic equations from E/IE. Their coefficients are specialised from a polynomial ring $A_0 = \mathbb{Z}[C_1, \ldots, C_M]$. This auxiliary ring has the proved constructive coherence, discreteness and noetherian properties needed internally. Transport of the final finite identities then gives the result over arbitrary R, without deciding equality or ideal membership in R.

Sections without decomposing schemes. Write J for the homogeneous relations and $I = (b_1, \ldots, b_t)$. The proof constructs cochains z_r by explicit geometric-series identities, with

$$dz_r \in I^{r+1}C^1 + JC^1, \qquad z_r - z_0 \in IC^0 + JC^0, \qquad (z_r)_Z = Z^{-1}.$$

These are the algebraic replacement for the zero-extended sections on W_{r+1}. The formulas do not require chart idempotents or a choice of a compatible infinite family of representatives.

One dependence relation instead of a uniform cohomology theorem. Over $D = A_0[U_1, \ldots, U_t]$, the evaluated Rees presentation uses the actual kernel of

$$D[X, Z] \longrightarrow (A_0[X, Z]/J)[T], \qquad U_j \longmapsto b_j T.$$

Its finite bihomogeneous generators and their J-certificates are constructed. Image powers of I are used, so no flatness assumption or identification of $I^r \otimes M$ with $I^r M$ is made.

The obstruction cochains are raised through a bounded partial free resolution. Monomial contractions leave only finitely many top Laurent coefficients, in a finite free module D^v. Each obstruction merely has such a partner. Noetherian induction obtains a dependence by using one partner at a time inside the existential conclusion. There is no choice of a whole sequence of partners.

After projection to one coefficient degree and evaluation $U_j = b_j$, this gives, for some N, coefficients $r_i \in I^{N-i}$ and $v \in I^{N+1}C^0$ such that

$$d\left(z_N - \sum_{i<N} r_i z_i - v\right) \in JC^1.$$

The corrected section still agrees with z_0 modulo $I + J$, since $N - i \geq 1$. This includes $N = 0$. Extraction gives (1).

Finite syzygy kernels and the Rees kernel are constructed in this argument. What is avoided is a separate computation of generators for the whole cohomology obstruction module, and a bound valid for all approximate sections. The theorem asks only for one dependence in the sequence at hand.

Assessment of originality. The ingredients—projective compactification, Rees finiteness, Čech contractions, dimension shifting, denominator clearing and the determinant trick—are established. The specific organisation around one sequence, partners under mere existence, and a single finite stable module is the feature worth documenting. The sources examined here do not present that combination. That observation supports describing the work as an *alternative constructive proof by finite section lifting*; it does not establish that the combination has never appeared elsewhere.

For perspective, Zariski's 1949 analytical proof uses valuations, completions of local domains and analytical irreducibility, notably in Theorems 3–4 [7]. It is a more distant comparison than the explicit section-lifting mechanism above. No exhaustive review of Zariski's other papers, EGA IV, or later textbooks was made here.

5. Connection with the Unit development

The mathematical proof is in `unit/doc/zmt-proof.tex`; the current endpoint and scope are recorded in `unit/sag/zmt-general/README.txt`. The correspondence is:

Mathematical step	Directories below `unit/sag/`
Finite residual data and arbitrary-base descent	`zmt-descent`
Explicit approximate sections	`zmt-sections-general`
Finite dependence and weighted projection	`zmt-certificate-general`, `zmt-weighted-general`
Rees membership and correction	`zmt-rees-general`, `zmt-correction-general`
Degree-lowering relations	`zmt-extract-general`
One stable module and determinant trick	`zmt-certificate-extract-general`
All finite numbers of generators	`zmt-general/FinalZMTGeneral.seg`

The public result is

`FinalZMTGeneral.Theorem[R,setR,I,propI,Ea,n,x,gen,fo].result`

It includes a finite spanning family for the actual image subalgebra $B \subseteq E$ and an equivalence given by the canonical localisation map. The noetherian auxiliary base and the dependence theorem are internal, not hypotheses of this endpoint. The development's ambient univalence, replacement and surjection parameters are unchanged; it introduces neither excluded middle nor a choice principle.

This comparison reads the proof and its recorded formalisation status; it does not report a new checker run. Its scope is the displayed finite-fibre algebraic theorem. A global quasi-finite scheme factorisation requires additional covering and gluing statements. The classical reconstruction in Section 3 is explanatory; it is not a separate Unit formalisation of schemes, Stein factorisation or formal functions.

References

[1] M. E. Alonso, T. Coquand and H. Lombardi, *Revisiting Zariski Main Theorem from a constructive point of view*, J. Algebra **406** (2014), 46–68. `https://arxiv.org/abs/1602.00597`. The local `ACL.pdf` was read, especially Theorem 1.3, Corollary 1.4 and §§2–3.

[2] A. Grothendieck, with J. Dieudonné, *Éléments de géométrie algébrique III, première partie*, Publ. Math. IHÉS **11** (1961), 5–167. `https://www.numdam.org/articles/10.1007/BF02684273/`. Original scan inspected at (4.1.7.12)–(4.1.7.21), pp. 127–129; (4.2.1), pp. 129–130; (4.3.1)–(4.3.3), pp. 130–131; (4.4.1)–(4.4.3), pp. 135–136. These are the volume page numbers, not the alternative continuous pagination printed at the foot.

[3] G. Horrocks, *Projective modules over an extension of a local ring*, Proc. London Math. Soc. (3) **14** (1964), 714–718. `https://doi.org/10.1112/plms/s3-14.4.714`. Theorem 1 and its proof were read in the local `phorrocks.pdf`.

[4] The Stacks Project, §10.123, *Zariski's Main Theorem*, Tag 00PI. `https://stacks.math.columbia.edu/tag/00PI`.

[5] The Stacks Project, §30.20, *The theorem on formal functions*, especially Lemmas 30.20.1 and 30.20.3. `https://stacks.math.columbia.edu/tag/02O7`.

[6] The Stacks Project, Lemma 30.20.4(3), Tag 02OB. `https://stacks.math.columbia.edu/tag/02OB`.

[7] O. Zariski, *A simple analytical proof of a fundamental property of birational transformations*, Proc. Nat. Acad. Sci. USA **35** (1949), 62–66. `https://doi.org/10.1073/pnas.35.1.62`. The paper was inspected through the NCBI scan/text.